Ref No: **Ex/EE/PC/B/T/221/2025**

B. E. ELECTRICAL ENGINEERING SECOND YEAR SECOND SEMESTER EXAMINATION - 2025

SUBJECT: - ELECTRICAL INSTRUMENTATION

Time: Three hours

Full Marks 100
(50 marks for each part)

Use a separate Answer-Script for each part

Question No.	PART I	Marks
	Answer Q.5 and any two questions from the rest.	
1.	Justify or correct ***any four*** of the following statements with suitable reasons in brief / derivations. (CO2-K2)	4×5=20
(a)	In capacitive level gauges for conducting liquids, the conducting liquid serves as one of the electrodes.	
(b)	The fringing effect of electric field lines in capacitive sensors can be minimized by opting for a push-pull (differential) configuration.	
(c)	In LVDTs, rectifier circuits may be deployed for getting an output replicating the pattern of variation of the displacement (being sensed) with time.	
(d)	Piezoelectric transducers are well-suited for measuring static mechanical force.	
(e)	In transit-time ultrasonic flowmeters in the wetted sensor variety, the differential transit time of the ultrasound is affected by the temperature of the liquid.	
2. (a)	Connections from a piezoelectric sensor are taken to a voltmeter via a coaxial cable 20 m long. The sensor material has a '*d*' constant of 150 pC/N. Capacitance of the sensor is 100 pF and the leakage resistance is 10^4 MΩ. Input capacitance of the voltmeter is 20 pF and its input resistance is 200 MΩ. The variety of coaxial cable available has a shunt capacitance of 10 pF/m. If the sensor experiences a force (in N) given by	

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Question No.	PART I	Marks
	$f(t)=4Sin(10t)$, where t is in second, obtain the expression for the voltage appearing across the voltmeter terminals under steady-state condition. Derive the expression(s) used. (CO2-K2)	10
(b)	What type of circuit would you propose for piezoelectric force sensors, in which the circuit parameters of the sensor, and the length of the connecting cable, would not affect the performance of the measurement? How can the performance of the system for lower frequency force signals be bettered? Give clear explanations. (CO2-K2)	10
3. (a)	"A logarithmic amplifier can be used for linearizing the output versus temperature characteristic for NTC thermistor based temperature measurement system."– Justify or correct this statement. Give suitable figure and derivations. *Practical example of any opamp based log circuit is not required.* (CO2-K2)	10
(b)	A temperature sensor has a nominal resistance of 1 kΩ at 25°C. Its resistances at 40°C and 100°C are 1124 Ω and 1706 Ω respectively. It is to be used in an ohmmeter circuit, over 40°C to 100°C How can you improve the linearity of the output voltage versus temperature characteristic by introducing an external resistance? Calculate the value of such resistance. Derive the expression used. (CO2-K2)	10
4.	Write short notes on *any two* of the following: (CO2-K2)	10+10
(a)	Ultrasonic flowmeter of non-wetted sensor variety.	
(b)	Level gauges for non-conducting liquids.	
(c)	Synchronous demodulator for capacitive sensors.	

5.	Answer any Two:	5+5
(a)	"In an astable multivibrator circuit used for variable air-gap capacitive sensors, the deviation of the output frequency from its nominal value is linearly related with the displacement being measured". Justify or correct the statement with necessary illustration and derivation. (CO5-K4)	
(b)	Show that the performance of a transformer double ratio bridge for capacitance measurement is inherently free from the effect of parasitic capacitances. (CO5-K4)	
(c)	"In an LVDT, the differential output voltage is in phase with the input voltage." – Comment whether the statement is true or fasle. Justify in favour of your argument. (CO5-K4)	

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PART -II

Time: Three Hours **Full Marks: 100**
(50 Marks for each part)

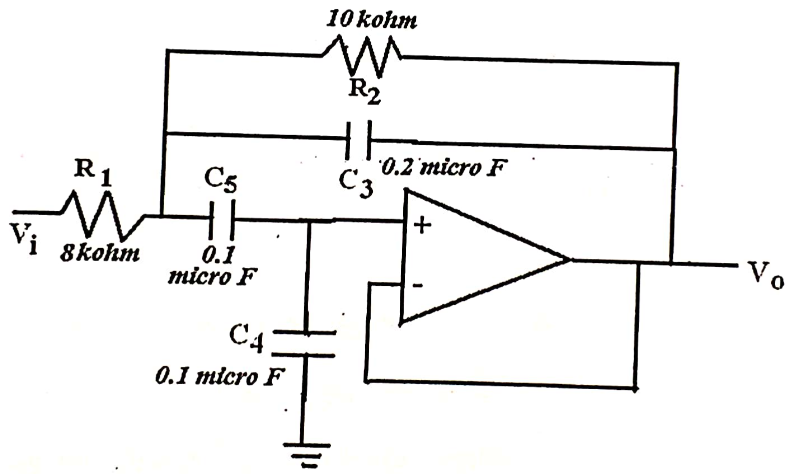
Use separate Answer-Script for each part
Answer *All* Questions

1. Explain the functions of "Trigger input" in a Cathode Ray Oscilloscope (CRO).
(CO1) [5]

OR

1. The input resistance and capacitance of a Cathode Ray Oscilloscope (CRO) are 2.0 μF and 1.0 MΩ. What is the matched capacitance of the probe for proper compensation assuming 100 kΩ probe resistance? Explain necessary formula.
(CO1) [5]

2.a) Draw the Switched capacitor implementation of the following circuit. Assume switching frequency is 1 kHz.



(CO3) [3]

b) Why State Vartiable Filters are called Universal Filter? How can you realize a 2nd order bandpass filter using state variable representation? Show necessary derivations.
(CO3) [2+8]

OR

2.a) Develop a linear model of Phase Locked Loop (PLL).
(CO3) [8]

b) How can a Phase Locked Loop (PLL) be used as frequency demodulator? (CO3) [5]

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3. a) Explain the operation of unipolar successive approximation type Analog to Digital Converter (ADC) with a flow chart and necessary diagram. (CO4) [7]

b) Obtain a 5-bit binary representation of an analog signal value of 10.317 V using successive approximation type ADC. Reference voltage is 16 V. (CO4) [4]

c) Comment on the quantization errors of truncation and rounding-off type ADCs. (CO4) [4]

OR

3 a) Explain with circuit diagram the operation of an R-2R ladder Digital to Analog Converter (DAC) considering 3 bits. (CO4) [9]

b) Explain linearity error of a DAC. (CO4) [4]

c) For a 4-bit DAC, the analog values corresponding to digital inputs 0000 and 0001 are 0.0125 V and 0.0625 V respectively. Find the output for an input 1110. (CO4) [2]

4. a) Show that the poles of Butterworth filter are located on an s-plane unit circle whereas Chebyshev poles are on an s-plane ellipse. (CO6) [6]

b) Find the transfer function of a 2nd order band pass filter circuit with center frequency $\omega_0 = 10$ krad/sec and bandwidth B = 2krad/sec and passband gain 10. Realize the band pass filter circuit. Also realize the band stop filter circuit from this band pass filter having same center frequency and pass band gain. (CO6) [9+2]

OR

4. a) Explain why the -3dB cut-off frequency of a normalized Chebyshev filter is greater than 1rad/sec.? (CO6) [4]

b) Find out the minimum order and ripple factor of a normalized Chebyshev lowpass filter with maximum passband ripple attenuation is 0.3 dB and stop band attenuation is minimum 40 dB for $\omega \geq 10$ rad/s. (CO6) [4]

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c) Design a low pass filter with following specifications using any of your known filter topologies:

- 3 dB cut-off frequency = 2 kHz.
- Pass-band attenuation is maximum 0.5 dB for $f \leq 1$ kHz.
- Stop-band attenuation is minimum 10 dB for $f \geq 5$ kHz
- Pass-band gain is 2.5

(CO6) [9]